

# Duy Nguyen

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## Education

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**National University of Singapore**, B.Comp. in Computer Science – Singapore Aug 2022 – present

- GPA: 4.86/5.00 ([Transcript](#)), concentrated in Algorithms & Theory and Artificial Intelligence
- Top Students for Discrete Structures, Data Structures and Algorithms, Programming Methodology, Software Engineering, Design and Analysis of Algorithms, Operating Systems, Competitive Programming

**National University of Singapore**, B.Sc. in Data Science and Analytics – Singapore Aug 2022 – present

- GPA: 4.86/5.00 ([Transcript](#)), concentrated in Operations Research, minored in Quantitative Finance

## Experience

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**Software Engineer Intern**, Citadel Securities – Hong Kong May 2026 – Aug 2026

kdb+/q, Python, C++, TypeScript

- Refactored a high-throughput streaming engine, migrating in-memory state to partitioned disk storage with local caching to eliminate critical memory bottlenecks. Engineered a streaming statistical service computing rolling metrics over high-frequency tick data to detect data divergence in real time with interactive dashboards for monitoring and alerting.

**Software Engineer Intern**, AI Centre for Educational Technologies – Singapore May 2025 – Aug 2025

TypeScript, React, Material UI, Tailwind CSS, Ruby on Rails, PostgreSQL, OpenAI API

- Enhanced [Coursemology](#), an open-source gamified LMS used by thousands of students and instructors. Integrated AI-assisted grading and programming plagiarism services, enabling automated rubric feedback and similarity scans to enhance grading consistency and reduce manual workload for instructors.

**Algorithm Engineer Intern**, Vilota – Singapore May 2024 – Aug 2024

C++, Python, Linux, OpenCL

- Integrated AprilTag 3 to accelerate fiducial marker detection, achieving 6x faster camera calibration. Implemented real-time panorama generation algorithm with parallel processing on i.MX 8M Plus Processors.

**Software Engineer Intern**, Computing for Voluntary Welfare Organisations – Singapore May 2023 – Aug 2023

TypeScript, React, Material UI, Golang, PostgreSQL

- Built multi-tenant admin dashboards and a low-code audit reporting engine for 400+ social workers in 11 Active-Aging Centres, reducing custom report turnaround time from days to hours.

## Projects

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**PeerPrep Interview Platform** Sep 2025 – Nov 2025

JavaScript, Node.js, MongoDB, Google Cloud Platform (GCP)

- Architected a scalable serverless microservices infrastructure on GCP with Infrastructure as Code (IaC) using Terraform.
- Engineered a collaboration service using WebSockets and CRDTs (Yjs) for conflict-free, real-time shared code editing.

**Address Element Extraction** Mar 2021

Python, Pytorch, Transformers

- Fine-tuned a named-entity recognition model to extract street names from Indonesia addresses given by Shopee users.
- Attained 1st place out of 1,000+ teams in the Data Science Challenge of Shopee Code League 2021.

## Skills

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**Full Stack Development:** Frontend (React, Material UI, Tailwind CSS), Backend (Node.js, Golang, Ruby on Rails, FastAPI, C++), Database (PostgreSQL, MongoDB, kdb+/q, Redis), DevOps (Linux, Git, Docker, CI/CD, Terraform, AWS, GCP)

**Data Science:** Programming (R, Python), Frameworks (PyTorch, Transformers), Tools (Weights & Biases, Jupyter, LaTeX)

## Awards

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- **Finalists** for various CTF events: DSTA Brainhack CDDC, GovTech Singapore AI CTF, NUS Greyhats Grey Cat The Flag
- **First Prize** for Computing Track at Huawei ICT Competition Asia Pacific 2023-2024
- **Silver Awards** for National Olympiad in Informatics (2019–2021) and Singapore Mathematical Olympiad (2020–2021)